Microeconomics I

Class 8 — Choice under uncertainty: expected utility and risk aversion

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Plan for today

Warm-up; two choices, by hands up

Lotteries, continuity, independence

Your two choices, revisited

Expected utility: form, theorem, uniqueness

Money lotteries and risk aversion

Certainty equivalent and risk premium

Break

Practice: five questions on Part I so far

Summary

Practicalities

- Problem set 2: handed in at the start of class, Wed 21 Oct; returned graded on Tue 27 Oct
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: always true, or not?

Only good 1 gets dearer, $p^0 \to p^1$. At p^0 she chose x^0, with utility u^0.

Fixed basket: $p^1 \cdot x^0 \,/\, p^0 \cdot x^0 = 1.3$. True cost-of-living index: $I = e(p^1, u^0) \,/\, e(p^0, u^0)$.

$I \leq 1.3$	$I \geq 1$
$I = 1.3$	good 1 inferior $\Rightarrow I > 1.3$

Class 6, slides 13, 22. Mas-Colell, Whinston and Green (MWG) Props. 3.E.1, 3.E.2, pp. 58–59.

Two choices: which would you take?

Imagine the prizes are real and paid once. No right answer.

First choice

A: 1 million euros for sure

B: 10 %: 5 million euros
89 %: 1 million euros
1 %: nothing

Second choice

C: 11 %: 1 million euros
89 %: nothing

D: 10 %: 5 million euros
90 %: nothing

Hands up: A or B? Then: C or D?

A classic pair of choices; the source in class.

Lotteries and expected utility

MWG 6.B

A simple lottery is a list of probabilities: a point in a triangle

Simple lottery: $L = (p_1, \ldots, p_N)$, $p_n \geq 0$, $\sum_n p_n = 1$; p_n = probability of outcome n.

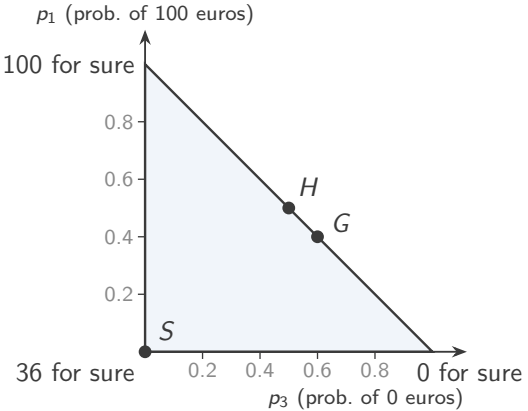
Outcomes: 100, 36, 0 euros.

$S = (0, 1, 0)$: 36 euros for sure

$H = (0.5, 0, 0.5)$: a coin, 100 or 0

$G = (0.4, 0, 0.6)$: 40 % chance of 100

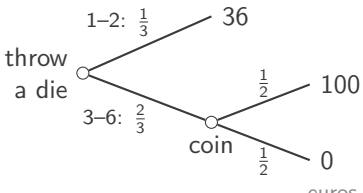
MWG Def. 6.B.1, p. 168; Fig. 6.B.1, p. 169 draws the same set as an equilateral triangle.



$p_2 = 1 - p_1 - p_3$: 0 on the hypotenuse, 1 at S

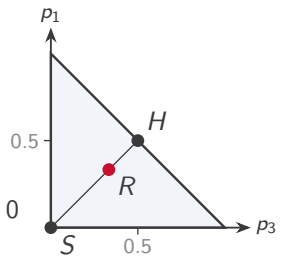
A compound lottery reduces to a simple one: $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

Compound lottery $(L_1, \ldots, L_K; \alpha_1, \ldots, \alpha_K)$: you get L_k with probability α_k.
Reduced lottery: $\alpha_1 L_1 + \cdots + \alpha_K L_K$.



$R = (S, H;\ \frac{1}{3}, \frac{2}{3})$

$\frac{1}{3}(0,1,0) + \frac{2}{3}(0.5,0,0.5)$
$= (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
= one throw: 1–2 → 100,
3–4 → 36, 5–6 → 0



Only the reduced lottery matters (MWG's consequentialist premise): the die-then-coin tree and one throw are the same lottery.

MWG Def. 6.B.2, pp. 169–170; Figs. 6.B.2–6.B.3, p. 170.

Rational and continuous preferences over lotteries

Rational: complete, transitive (class 1).

Continuous: for all $L, L', L'' \in \mathcal{L}$ the sets $\{\alpha \in [0,1] : \alpha L + (1-\alpha)L' \succsim L''\}$ and $\{\alpha \in [0,1] : L'' \succsim \alpha L + (1-\alpha)L'\}$ are closed.

Small probability changes never reverse a strict ranking.

> **Fails: safety first** (no chance of 0 beats any chance).
> $L = 100$, $L' = 0$, $L'' = 36$ euros for sure: second set $= [0, 1)$.

MWG Def. 6.B.3, p. 171 ("safety first": same page). Class 3: lexicographic order, not continuous.

Independence: mixing both with the same L'' keeps their ranking

Independence axiom: for all $L, L', L'' \in \mathcal{L}$ and $\alpha \in (0,1)$:

$$L \succsim L' \iff \alpha L + (1-\alpha)L'' \succsim \alpha L' + (1-\alpha)L''.$$



$\alpha L + (1-\alpha)L''$ $\qquad$ $\alpha L' + (1-\alpha)L''$

Tails gives both the same L'', and you get it instead of L or L', never with it: unlike bundles of goods.

MWG Def. 6.B.4, p. 171 (fn. 5: von Neumann and Morgenstern, 1944); coin-toss reading and Fig. 6.B.4, p. 172.

Revealed in class

Before class: in each pair, what do the two lotteries have in common?

Expected utility: $U(L) = u_1 p_1 + \cdots + u_N p_N$ is linear in probabilities

U has the expected utility form if there are numbers $u_1, \ldots, u_N$ with $U(L) = u_1 p_1 + \cdots + u_N p_N$ for every L.

Such a U is a von Neumann–Morgenstern (vNM) expected utility function.

Her utilities for 100, 36, 0 euros: 10, 6, 0.

$U(S) = 6, \quad U(H) = 5$

$U(R) = \frac{1}{3} \cdot 10 + \frac{1}{3} \cdot 6 + \frac{1}{3} \cdot 0 = \frac{16}{3} = \frac{1}{3}\, U(S) + \frac{2}{3}\, U(H)$

MWG Def. 6.B.5 and Prop. 6.B.1 (expected utility form $\iff$ U linear), p. 173.

Continuity + independence ⇒ expected utility: straight, parallel lines

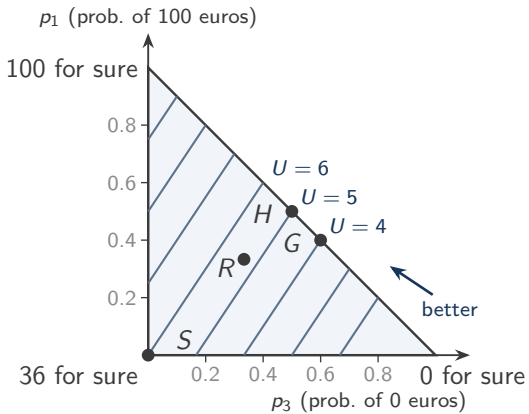
Expected utility theorem: if $\succsim$ on $\mathcal{L}$ is rational, continuous and satisfies independence, there are $u_1, \ldots, u_N$ such that for all L, L':

$$L \succsim L' \iff \sum_n u_n p_n \geq \sum_n u_n p'_n.$$

In the triangle: indifference curves are straight, parallel lines.

Conversely, every expected utility U satisfies independence (MWG Ex. 6.B.2).

MWG Prop. 6.B.3, (6.B.4), p. 176; proof pp. 176–178; Fig. 6.B.5, p. 175.



Expected utility is unique up to $\beta U + \gamma$, $\beta > 0$; u^2 flips her choice

utilities of (100, 36, 0 euros)	$U(G)$ $G = 40\,\%$ of 100	$U(S)$ $S = 36$ for sure	she picks
$u = (10, 6, 0)$	4	6	S
$2u + 3 = (23, 15, 3)$	11	15	S
$u^2 = (100, 36, 0)$	40	36	G

$\tilde{U}$ is another vNM utility for $\succsim \iff \tilde{U} = \beta U + \gamma$ with $\beta > 0$.

u^2 keeps the order of prizes, not lotteries:
unlike class 3, expected utility is cardinal.

MWG Prop. 6.B.2, pp. 173–174.

Money lotteries and risk aversion

MWG 6.C

A money lottery is a distribution function F; $U(F) = \int u(x)\, dF(x)$

$F(x)$ = probability that the payoff is at most x.

$U(F) = \int u(x)\, dF(x)$; with prizes x_n and probabilities p_n: $\sum_n p_n\, u(x_n)$.

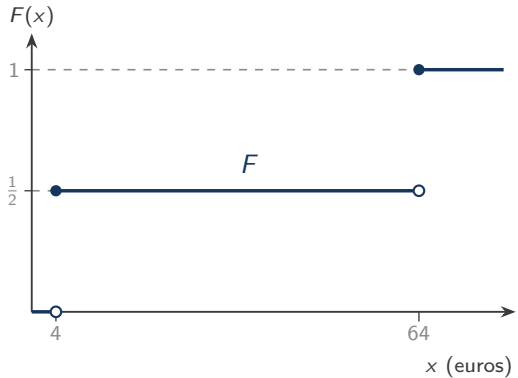
U: vNM utility, on lotteries.

u: Bernoulli utility, on money — increasing, continuous.

Her 10, 6, 0 for 100, 36, 0 euros: $u(x) = \sqrt{x}$.

MWG p. 183; (6.C.1) and fn. 12, p. 184; p. 185.

F: 4 or 64 euros, $\frac{1}{2}$ each



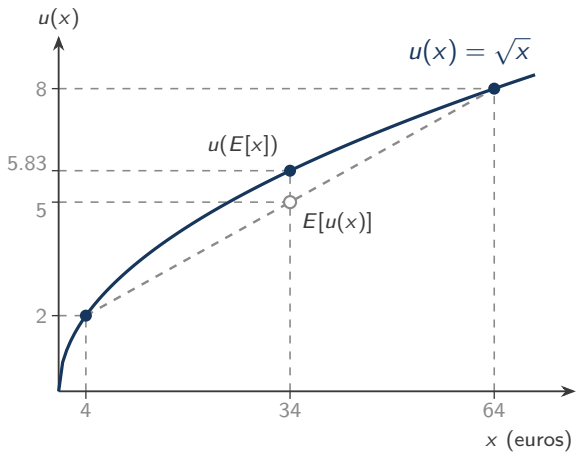
Risk averse $\Leftrightarrow$ u concave: the chord lies below the graph

Risk averse: for every F, the sure amount $\int x\, dF(x)$ is at least as good as F.

With expected utility: $\int u(x)\, dF(x) \leq u(\int x\, dF(x))$ for every F — Jensen: u concave.

Concave in money: a property of the function u, not convexity of preferences over bundles (classes 2–3). U is linear in probabilities.

MWG Def. 6.C.1 and (6.C.2), p. 185; Fig. 6.C.2, p. 186.



Certainty equivalent 25 euros, risk premium 9 euros: three steps

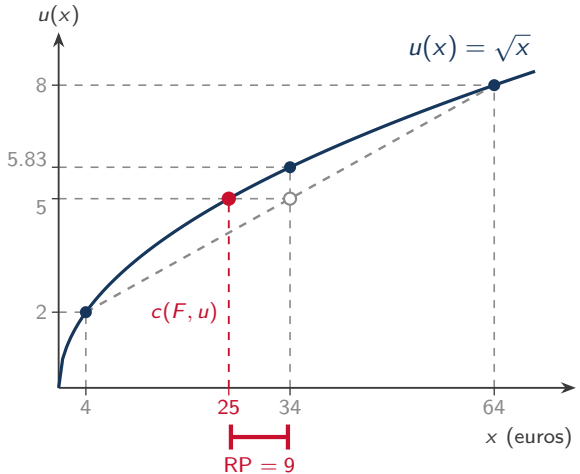
Certainty equivalent $c(F, u)$:

$u\Big(c(F,u)\Big) = \int u(x)\, dF(x).$

Risk premium: $\mathrm{RP} = E[x] - c(F,u)$.

1. $E[x] = \frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 64 = 34$ euros
2. $E[u(x)] = \frac{1}{2} \cdot 2 + \frac{1}{2} \cdot 8 = 5$ utils
3. $c(F,u) = u^{-1}(5) = 25$ euros;
   $\mathrm{RP} = 34 - 25 = 9$ euros

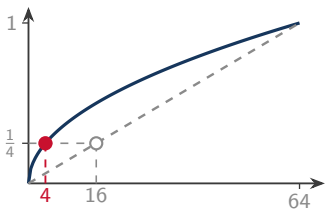
MWG Def. 6.C.2(i), (6.C.3), p. 186: "some expected return is traded for certainty". RP as in Muñoz-Garcia, p. 357.



Concave u: $c(F,u) \le E[x]$ (MWG Prop. 6.C.1); convex u: $c(F,u) \ge E[x]$

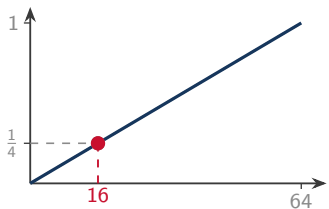
Toss two coins. Both heads: 64 euros; otherwise nothing. $E[x] = 16$ euros for everyone.

Ana: $u = \sqrt{x}/8$



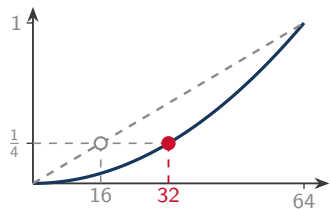
$c = 4 < 16$

Ben: $u = x/64$



$c = 16$

Carla: $u = x^2/4096$



$c = 32 > 16$

Each u rescaled so that $u(0) = 0$, $u(64) = 1$ (Prop. 6.B.2): $E[u(x)] = \frac{1}{4}$ in all three.

Risk averse $\iff$ u concave $\iff$ $c(F,u) \le E[x]$ for every F, i.e. RP ≥ 0.

MWG Prop. 6.C.1, p. 187.

Break

Ten minutes

Practice

Five questions on Part I so far: try first, then we solve them together

Practice 1: two dice games

Prizes 150, 60, 0 euros; a lottery lists their probabilities in this order.

Iris's preferences over lotteries satisfy the independence axiom. Throw a fair die:

Game X: on 1–3, 60 euros for sure; on 4–6, the lottery $M = (0,\ \frac{1}{2},\ \frac{1}{2})$.

Game Y: on 1–3, a fair coin for 150 or 0 euros; on 4–6, M.

Iris strictly prefers Y to X.

(a) Reduce X and Y.

(b) Which must be true?

A She strictly prefers 60 euros for sure to the coin.

B She is risk loving.

C She strictly prefers the coin to 60 euros for sure.

D Nothing follows unless we know how much she values 150, 60 and 0 euros.

Answers revealed in class.

Practice 2: three equally likely prizes

A lottery F pays 0, 9 or 36 euros, each with probability $\frac{1}{3}$.
Rosa is an expected utility maximizer with Bernoulli utility $u(x) = \sqrt{x}$.

(a) $E[x]$ and $E[u(x)]$.

(b) Her certainty equivalent $c(F, u)$ and the risk premium $\text{RP} = E[x] - c(F, u)$.

(c) Instead of F she is offered 12 euros for sure. Does she accept? Would a risk-neutral person accept?

Answers revealed in class.

Practice 3: what duality says

Vera's preferences are rational, continuous, locally non-satiated and strictly convex. At prices $p = (2, 7)$, the cheapest bundle that gives her utility 5 is $h(p, 5) = (7, 4)$.

h: Hicksian demand; x: Walrasian demand.

Which statement must be true?

A $h((4, 14),\ 5) = (14, 8)$

B $x(p, 42) = (7, 4)$

C $x(p, 84) = (14, 8)$

D $x(p, 40) = (7, 4)$

Answers revealed in class.

Practice 4: one price rise, four measures

Kim: $u(x_1, x_2) = x_1 x_2$, wealth $w = 18$.
Good 1 gets dearer: prices $p^0 = (1, 3) \to p^1 = (9, 3)$.

Class 4: $x(p, w) = \big(w/(2p_1),\ w/(2p_2)\big)$. h: Hicksian demand; e: expenditure function.

(a) $x^0 = x(p^0, w)$, u^0; $x^1 = x(p^1, w)$, u^1.

(b) The Hicksian bundle $h(p^1, u^0)$; on good 1, the substitution effect $h_1(p^1, u^0) - x_1^0$ and the wealth effect $x_1^1 - h_1(p^1, u^0)$.

(c) Compensating variation $\text{CV} = w - e(p^1, u^0)$ and equivalent variation $\text{EV} = e(p^0, u^1) - w$.

(d) The true cost-of-living index $e(p^1, u^0)/w$ and the fixed basket $(p^1 \cdot x^0)/(p^0 \cdot x^0)$.

Answers revealed in class.

Practice 5: a straight-line Bernoulli utility

Otto is an expected utility maximizer with Bernoulli utility $u(x) = 3x + 4$ over money $x \geq 0$, in euros.

Which statement is true?

A He is risk averse, since $u(0) = 4 > 0$.

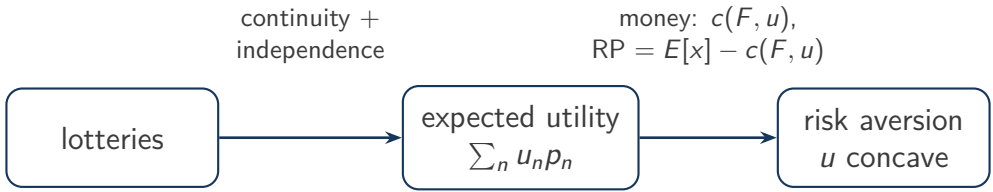
B His certainty equivalent of every lottery is 4 euros above its expected value.

C He strictly prefers 10 euros for sure to a fair coin for 0 or 20 euros.

D His certainty equivalent of every lottery equals its expected value.

Answers revealed in class.

Today in one line



Tuesday 27 October: Uncertainty II: Arrow–Pratt measures, comparing risk aversion, insurance. First- and second-order stochastic dominance, mean-preserving spreads. Review.

Read MWG 6.C–6.D (or Muñoz-Garcia ch. 5) for Tuesday 27 October.

Part I exam: Wednesday 28 October, Room 1030.
It runs from 09:30 to 11:20.